

Dataset and Source Evaluation

1. MedQuAD dataset:
 - Main source: National Institutes of Health (NIH)
 - Has more than 40,000 QA elements
 - HIPAA compliant (has no patient information)
 - Information comes directly from NIH, not from peer-reviewed research journals
 - Focuses on general consumer health topics
2. PubMedQA:
 - Its based on PubMed article Abstracts
 - Answer mainly yes/no/maybe questions
 - HIPAA compliant
 - All sources are peer-reviewed
3. HealthCareMagic-100k:
 - Contains patient-doctor QA
 - Is in conversational format
 - HIPAA Compliant
 - The content is not peer-reviewed, but user generated
 - Does not contain sources to cite.
4. MedDialog:
 - Contains large-scale doctor-patient conversations
 - Not originally in English
 - Good conversational flow
 - HIPAA complaint (conversations are de-identified, but we should still check just in case)
 - Does not contain sources to cite, only conversations
5. MedAESQA:
 - Contains medical questions with manually annotated answers
 - HIPAA compliant
 - Every answer contains a citation to a peer-reviewed article
 - Mostly public health queries
 - Relatively small dataset

Note: for better performance and minimize hallucinations we should also use RAG over a collection peer-reviewed articles like (PMC FTP bulk download).